DressMe BO7

Virtual Try-On Final Notebook Report

Research, benchmarking, runtime optimization and wardrobe validation

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Executive summary

The exported notebook evidence establishes a working pretrained virtual try-on baseline. FASHN VTON v1.5 was selected with **float16, 15 sampling steps, guidance 1.5 and seed 42**. Measured median latency was 34.7 seconds on six changed-garment cases and 35.9 seconds on eight BO6 catalogue garments, using one Kaggle Tesla T4.

Two candidates were compared on 12 frozen cases: six paired reconstructions and six garment replacements. All **36 main benchmark runs** succeeded across FASHN and two CatVTON resolutions. Visual results show different failure patterns and do not establish a decisive quality winner. FASHN was selected mainly because it does not require a supplied person mask.

The contribution is the dataset audit, controlled inference comparison, hardware diagnosis, precision and sampling optimization, failure analysis, BO6 garment demonstration, and inference packaging. **No training or LoRA result is evidenced.** The artifacts use upstream pretrained weights; they contain no newly trained model.

Outcome	Evidence-based status
Pretrained VTON generation	Executed and evidenced in exported logs and images
Selected practical configuration	FASHN v1.5 / float16 / 15 steps
Research milestone	Completed with evaluation and reproducibility limitations
Complete BO7 product	Incomplete: application integration and mandatory Hybrid Avatar remain outside this evidence

Basis and access limits

This report summarizes the supplied research evidence ZIP, two inference bundles and prior research PDF. The actual .ipynb was not attached, so cell ordering, complete notebook code and the reported trainer cannot be audited directly. Exported results are inspected here; GPU inference was not rerun in this review. Evidence labels [E1]-[E4] are defined on page 8.

1. Objectives, data and preparation

The research objective was to choose a feasible pretrained model and settings for generating a visual simulation of a person wearing a BO6 wardrobe garment. The approach was iterative: audit inputs, establish baselines, diagnose failures, optimize measured bottlenecks, test wardrobe garments, and export the selected configuration.

BO6 catalogue audit [E1]

Check	Recorded result
Canonical image tree	44,441 images; another 44,441 files outside the selected tree
Metadata	44,424 rows; 22 malformed records quarantined
Integrity	0 corrupt images; 0 duplicate image-ID rows; 0 duplicate metadata-ID rows
Join mismatches	5 metadata rows without images; 22 images without metadata
Exact image-content duplication	1,431 rows belonging to duplicate-content groups
Initial VTON inventory	15,354 candidates: 12,515 tops; 2,384 bottoms; 455 one-pieces

The mirrored image tree was excluded using the canonical metadata location. The mirror report does not document a complete content-by-content comparison of the excluded tree. Duplicate-content rows are distinct from duplicate IDs: different catalogue IDs can share the same image.

One candidate is in Accessories rather than Apparel. An Apparel-only filter would retain **15,353 candidates**. This is a recommended integration filter, not a claim that the exported candidate CSV was already corrected.

VITON-HD preparation [E1]

The inspected paired inventory contains 4,000 tops cases, sampled from the official train/test layout. Binary masks were derived from original person photos and agnostic-v3.2 images: grey fill 128 +/- 14 near pixels changing by more than 24, followed by a median filter and dilation. Provenance records 3,926 masks created, 74 reused and zero recorded problems.

The split report records 256 training, 24 validation and six paired test cases; eight cross-official-split cases were quarantined. The final test manifest contains **12 rows**, combining paired and transfer cases. The 12-case development benchmark is the source of the main results; a separate final held-out evaluation is not evidenced.

All 4,000 person identities are unknown in the split report. Content grouping reduces exact duplication but does not prove identity separation, removal of near duplicates, or independence from upstream pretraining. Training/validation manifests were prepared even though adaptation was not executed.

2. Experimental protocol and runtime work

Dimension	Recorded setup
Hardware	Kaggle session with two Tesla T4 devices; workers use cuda:0
Runtime	Python 3.13.15; PyTorch 2.11.0+cu128; CUDA 12.8
FASHN source	7c0f10af3f91ad4048fe9729c470a13ef905d25a
CatVTON source	7818397f25613beedb3d861a34769f607cfcf3b1
CatVTON checkpoint	mix-48k-1024; base, attention and VAE snapshot revisions recorded
Cases and seed	Same 12-case development manifest; seed 42
Evaluation	Run success, inference time, output dimensions, PyTorch GPU memory and visual review

The manifest SHA-256 matches model_selection.json. Six paired cases already show the person wearing the target garment; FASHN can see that garment in its maskless input. These are useful reconstruction checks but can favour a maskless model. The six changed-garment cases better represent the intended product.

Runtime diagnosis and optimization

ONNX provider repair. The recorded FASHN runtime uses onnxruntime-gpu 1.26.0 with CUDAExecutionProvider in both detector and pose sessions. The earlier report attributes the initial CPU fallback to a CUDA library mismatch. The workers fail explicitly if CUDA is unavailable, preventing silent CPU benchmarking.

Precision correction. The runtime confirms no native BF16 support on this T4. The worker overrides the model to float16. The recorded official-sample BF16 run took 447.91 s; the float16 diagnostic took 65.49 s, a **6.84x latency reduction**. Recorded peak allocated memory fell from about 10.90 to 2.96 GiB. This comparison is one diagnostic sample, not a broad quality-equivalence study.

Resolution correction. CatVTON was evaluated at 384 x 512 and 768 x 1024. The higher setting improved the reported logo, sleeve and boundary failures, at higher compute cost. These are configuration experiments, not trained-model improvements.

Sampling reduction. FASHN was reduced from 30 to 20 and 15 steps. The 15-step choice was checked on two sweep cases, then six transfer cases and eight catalogue garments. CatVTON also received resolution/step sweeps.

The previous PDF describes setup failures, mask derivation, a CatVTON safety-checker false positive and photo-type correction. Where exact notebook execution is unavailable, these explanations remain reported work history. The exported workers confirm a benchmark configuration with the CatVTON safety checker disabled. [E2-E4]

3. Quantitative results

Main development benchmark: 36 successful runs [E1]

Configuration	Cases	Median s	Allocated GiB	Reserved GiB
FASHN / 30 steps / 576 x 768	12	71.44	2.96	5.08
CatVTON / 30 steps / 384 x 512	12	7.46	2.80	3.26
CatVTON / 30 steps / 768 x 1024	12	73.50	3.57	4.69

These configurations do **not** have matched output resolution. Similar latency for high-resolution CatVTON and FASHN is an observed practical tradeoff, not a same-resolution architecture comparison. Low-resolution CatVTON is about 9.6x faster than FASHN in this table, with visibly weaker quality reported in the prior review.

Speed sweeps and follow-up runs [E1]

Configuration / set	Cases	Median seconds
FASHN / 20 steps / sweep	2	45.80
FASHN / 15 steps / sweep	2	36.64
FASHN / 15 steps / changed garments	6	34.70
FASHN / 15 steps / BO6 corrected photo types	8	35.86
CatVTON / 20 steps / 768 x 1024	2	49.88
CatVTON / 30 steps / 576 x 768	2	26.06
CatVTON / 20 steps / 576 x 768	2	17.10

FASHN inference remains around 35-37 s in the selected configuration. The frequently quoted approximately 12x gain compares the 447.91 s diagnostic with the 36.64 s two-case sweep (12.23x). It combines precision/settings changes and different inputs; it is an **illustrative operational comparison**, not a controlled 12x experiment on a common case set.

Measurement boundaries

Latency is a GPU inference-pipeline measurement. Model initialization is logged separately. FASHN excludes image decoding; CatVTON excludes supplied-mask generation. Neither measurement represents upload, job waiting, persistence or full website response time. Peaks are PyTorch allocation/reservation values, not total device usage including ONNX sessions and driver overhead.

Run success means an image was produced; it does not mean the garment was reproduced faithfully. Two review CSVs contain identifiers but no completed quality scores. No numeric garment-fidelity score, independent acceptance rate or large-sample FID/KID result is available.

4. Visual results and wardrobe demonstration

Figure 1 shows the saved 30-step and 15-step transfer outputs. Broad appearance is similar in most cases, but the cream camisole fails at both settings: straps/silhouette change and unrelated trousers are repainted. Lower latency does not resolve that failure.

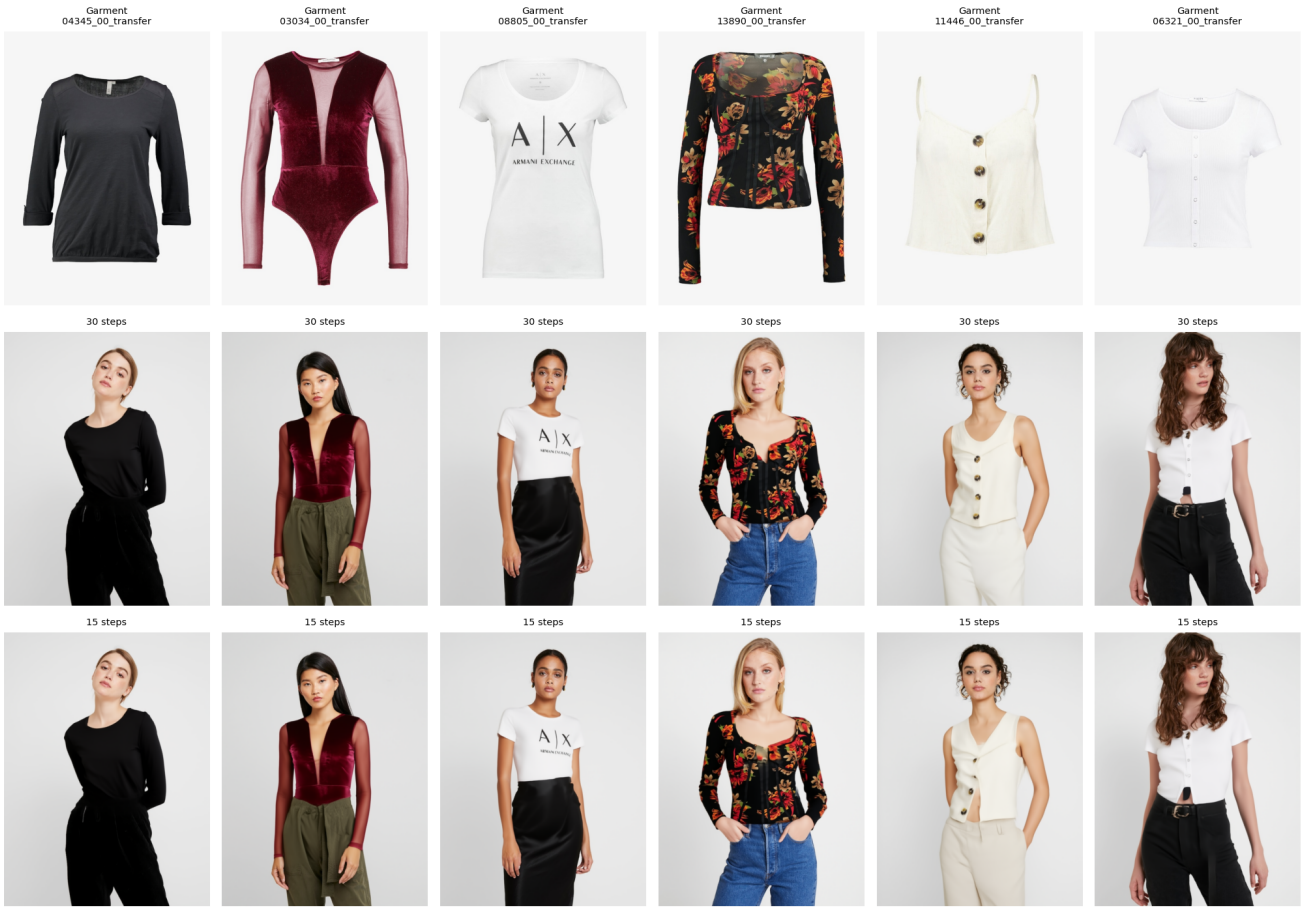


Figure 1. Garment inputs, FASHN at 30 steps, and FASHN at 15 steps. Source: exported transfer6 contact sheet [E1].



Figure 2. One VITON-HD person with eight BO6 catalogue garments, using FASHN at 15 steps. Source: corrected-photo-type demonstration [E1].

All eight corrected BO6 runs produced images. The prior PDF judges seven usable; that is an informal small-sample observation, not a validated 87.5% accuracy. The saved sheet visibly shows printed-detail limitations and the white top transferring blue jeans from the catalogue model. This demonstrates catalogue compatibility, with one benchmark person rather than a diverse real-user study.

5. Failure analysis and model decision

Observed issue	Evidence / interpretation	Product implication
CatVTON at low resolution	Reported missing sleeve, damaged logo and cropped hems; higher resolution improves results	Evaluate resolution before adaptation
CatVTON mask boundaries	Reported waistline truncation and clothing remnants	Mask quality and alignment need dedicated validation
FASHN unrelated changes	Cream camisole repaints trousers; white BO6 top imports catalogue jeans	Inspect preservation of non-target clothing
Wrong garment photo type	Orange flat-lay shirt initially treated as model-worn in prior report	Store and validate flat-lay versus model-worn
Text and print fidelity	Small garment text distorted in catalogue outputs	Do not promise exact logo or text reproduction

Why FASHN was selected [E1-E2]

FASHN v1.5 offers the simpler product path: person image + garment image + category + photo type, with no supplied person mask. Its reviewed quality was competitive with high-resolution CatVTON in the six transfer cases, and its selected latency was practical for an asynchronous demonstration. The decision is a justified engineering choice for this milestone, not proof of universal model superiority.

The selected settings are float16, 15 timesteps, one sample, guidance scale 1.5, seed 42 and segmentation_free=true. The alternative 576 x 768 CatVTON configuration at 20 steps took 17.10 s on two cases, excluding mask creation. The supplied CatVTON bundle still defaults to **768 x 1024 with 30 steps**; it does not default to that fast alternative.

Why no LoRA result is claimed

Training hyperparameters are present in experiment_config.json, including rank 4 and learning rate 0.0001. Their presence is preparation, not execution evidence. No adapter artifact, training-loss record or before/after trained-model comparison is supplied. The prior report states that the proposed trainer was never run; its code cannot be confirmed without the notebook.

Deferring adaptation was reasonable: the largest demonstrated improvements came from precision, resolution, sampling and input interpretation. Whether LoRA could improve remaining fidelity is unresolved. It should be tested only after a defined failure target, suitable supervision and an independent evaluation are established.

Quality-review limitations

The prior PDF describes two reviewers and a roughly tied comparison. model_selection.json records a single-reviewer judgement, and neither supplied review sheet has scores or reviewer records. This report retains the qualitative observations but does not claim a formally documented two-reviewer study. [E1, E4]

6. Packaging and reproducibility review

Both inference bundles include workers, restore code, model cards, configuration, requirements, licence text and SHA256SUMS.json. All **13 listed bundle-file checksums passed in each archive**. This establishes integrity of the supplied files; it does not establish successful restoration or GPU inference on a fresh machine. [E2-E3]

Important discrepancy in the FASHN export

The supplied FASHN model card contains **13 weight_checksums entries**: three model/ONNX payloads and ten files under .cache. restore_bundle.py validates every entry. Cache metadata can change between downloads, creating a restoration failure unrelated to the actual weights. The prior PDF says a rebuilt bundle removed these cache entries, but that correction is **not present in this attachment**.

Before a clean restore, restrict the weight manifest to model.safetensors, dwpose/dw-ll_ucoco_384.onnx and dwpose/yolox_l.onnx, then regenerate the model-card file checksum in SHA256SUMS.json. This report documents the corrective action; the supplied archives were not modified.

Reproduction procedure to validate next

Step	Required action / acceptance evidence
1. Prepare environment	Use recorded Python/Torch/CUDA versions and separate candidate environments. Preserve Torch constraints; install model-specific requirements and worker dependencies.
2. Correct and restore	Fix the FASHN cache checksum issue, run restore_bundle.py and verify source revision and actual weight hashes.
3. Supply inputs	Fill the manifest with local person/garment paths, category and photo type. CatVTON additionally requires an aligned binary mask.
4. Execute worker	Run the appropriate worker; inspect runtime provider, precision, output file and successful JSONL record.
5. Validate clean environment	Perform real inference after a fresh restore and compare a frozen reference case. Record image, latency and memory.

FASHN source is fixed by commit and payload hashes are recorded, but the restore script downloads weights through the upstream download script rather than a pinned weight repository revision. A future payload change could be detected without automatically retrieving the historical payload. CatVTON checkpoint snapshot revisions are pinned. Requirements alone may not be a complete worker environment; the notebook setup/pip logs remain relevant.

Licensing is recorded from the supplied artifacts: CatVTON is labelled CC BY-NC-SA 4.0 and the dataset mirror is described as CC BY-NC 4.0. FASHN has bundled Apache 2.0 code terms. Full checkpoint/dependency permissions were not reverified as a current legal audit. [E1-E4]

7. Final status, next milestone and evidence

Requirement / deliverable	Final status
Pretrained virtual try-on baseline	Executed; logs and sample outputs available
Model comparison and runtime tuning	Executed on small development sets; decision justified within that scope
BO6 wardrobe compatibility	Eight-garment notebook demonstration; application contract/integration unverified
LoRA / newly trained weights	Not executed / not evidenced
Clean reproducible export	Partially established: bundle hashes pass; FASHN cache issue and clean restore remain
Backend, frontend, private persistence, deployment	Not evidenced by the supplied research artifacts
Mandatory Hybrid Avatar	Outside this VTON work; no evidence here of automatic initialization, manual correction and saved product use

Conclusion

The VTON research milestone has produced a measured, useful baseline and a defensible configuration choice. Its strongest contribution is practical engineering: identifying a hardware precision bottleneck, comparing pretrained candidates, diagnosing input and resolution failures, and translating those findings into a wardrobe demonstration. The evidence supports moving toward implementation while retaining the documented quality limits.

This is a final report of the research exports, not a declaration that all BO7 work is complete. No physical-fit, size or fabric-behaviour guarantee is established by these generated images.

Smallest correct next action

Correct the FASHN export and run one clean restore/inference smoke check with a frozen person/garment pair. Acceptance evidence: fixed three-payload hash manifest, recorded source revision, CUDA pose providers, float16 precision, successful image and run log. Then integrate that worker with job status/errors, BO6 garment metadata and private storage. Hybrid Avatar remains a separate mandatory milestone.

Source and evidence index

ID	Supplied source / records used
E1	bo7_research_evidence (1).zip: baseline_runs.csv; runs/*.jsonl; audit/split/model-selection JSON; runtime; manifests; masks/provenance; visual sheets and diagnostic logs.
E2	fashn_inference_bundle (1).zip: worker, restore script, requirements, model card, configuration, licence and checksums.
E3	catvton_base_inference_bundle (1).zip: worker/common loader, checkpoint revisions, configuration, licence and checksums.
E4	DressMe BO7 - Virtual Try-On Research Report.pdf: prior narrative work history and qualitative judgements, checked against current exports.

Review method: archive extraction, JSON/CSV/log inspection, recalculation of run counts and medians, visual inspection of saved contact sheets, source-code inspection of bundle restoration/workers, and checksum validation. No notebook execution or fresh GPU restoration was performed.